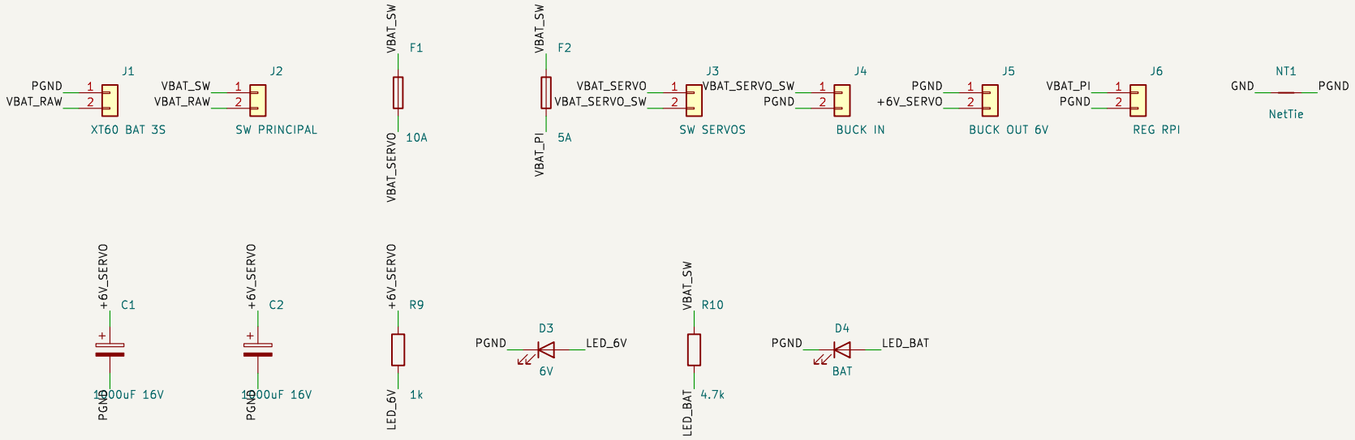
ALIMENTACION: Bateria 3S -> J2 interruptor general -> F1 (10 A) -> J3 interruptor servos -> J4/J5 buck 6 V -> servos
-> F2 (5 A) -> J6 regulador 5.1 V / 5 A con salida USB-C a la Raspberry Pi 5

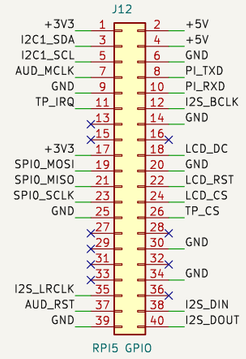
PGND (potencia) y GND (logica) se unen SOLO en NT1, junto a J1.

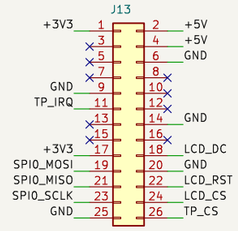


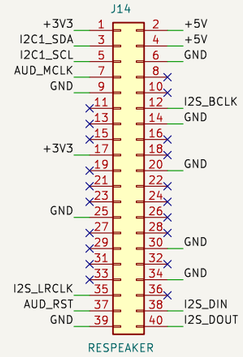
RASPBERRY PI 5 (cable plano 40 vias), PANTALLA Display-G 26 pines y ReSpeaker 2-Mics HAT V2.0

J13 y J14 replican la numeracion fisica de la Pi; solo se cablean los pines indicados.

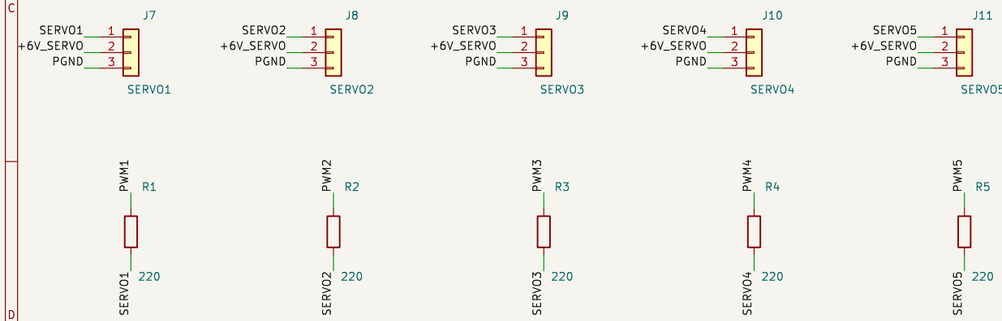
ReSpeaker: pines 8, 10, 11, 19, 23, 32, 33 y libres sin conectar.





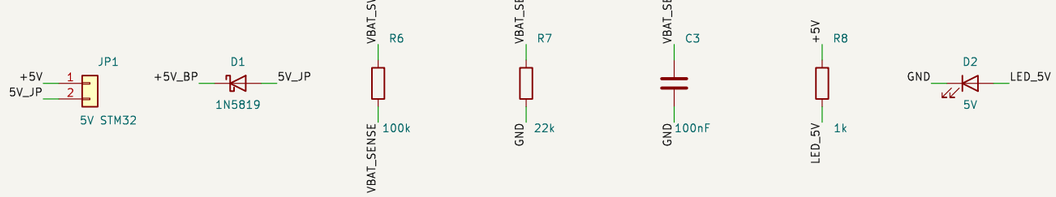


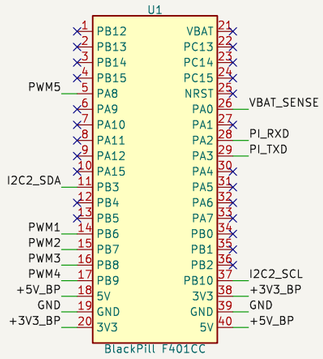
SERVOMOTORES MG996R (S, V+, GND) a 6 V. Senal PWM desde la Black Pill con 220 ohm en serie.



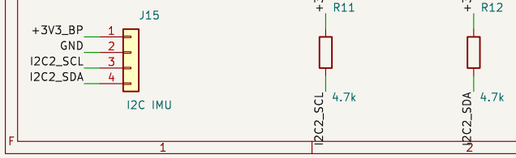
STM32F401 BLACK PILL: UART2 PA2 TX -> Pi RX (pin 10), PA3 RX <- Pi TX (pin 8). 5 V desde la Pi por JP1 + D1.

La 3.3 V de la Black Pill (+3V3_BP) NO se une a la 3.3 V de la Pi. PA0 mide la bateria: Vbat * 22k / 122k.









I2C2 para IMU futura (PB10 SCL, PB3 SDA).

Generado desde scripts/netlist.py

Raspberry Pi 5 + STM32F401 Black Pill + Display-G 3.5in + ReSpeaker 2-Mics + 5x MG996R

Tec de Monterrey — Campus Guadalajara

Sheet: /

File: cuty_carrier.kicad_sch

Title: C.U.T.Y — Placa base (carrier)

Size: A3	Date: 2026-11-02	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1